IT5006 FUNDAMENTALS OF DATA ANALYTICS · GROUP 10

Catch the bad review before it's written

Predicting late deliveries and unhappy customers on Brazil's Olist marketplace — and acting at the right moment

Checkout — Promised date — Delivery — Review

Pan Pinyou · Li Yudan · Zhu Gengyu

National University of Singapore · November 2026

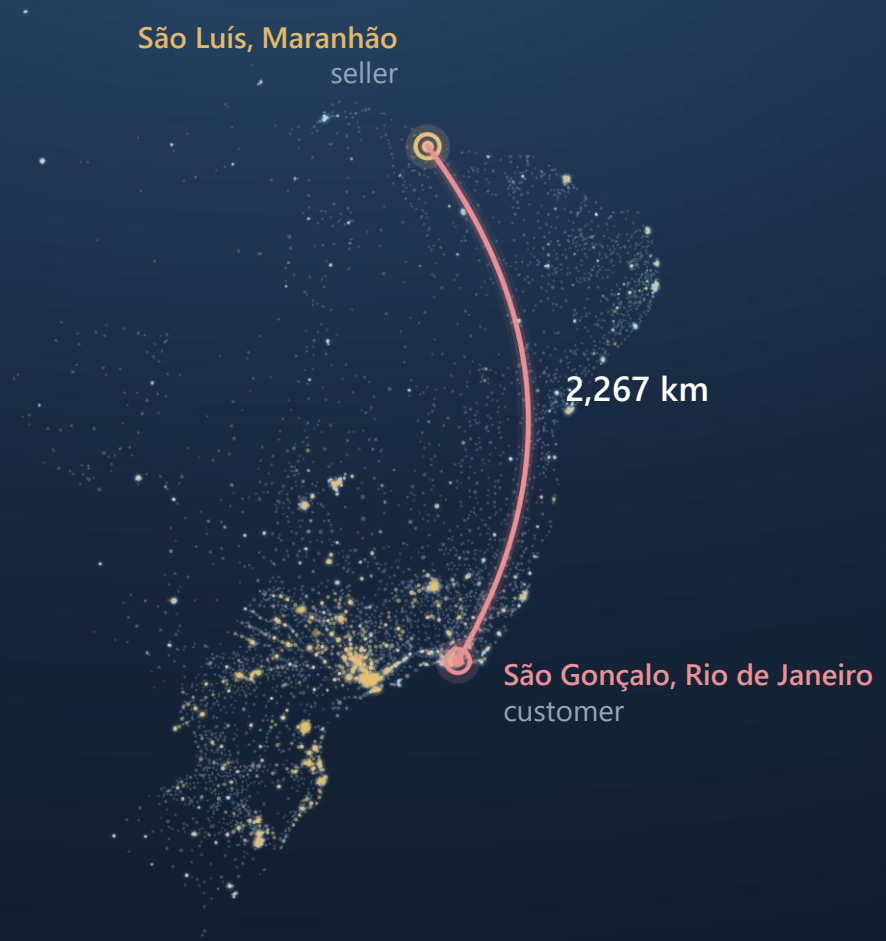


try the live app

Each dot is a postcode with Olist customers (blue) or sellers (gold)

This review arrived before the parcel did

Order e1320973 · health & beauty, R$65 + R$28 freight · never seen by the models during training



16 Jul day 0
Bought. Promised by 7 August (21 days).

7 Aug day 22
Promised date passes. The parcel is still on the road.

9 Aug day 24
Olist emails the review survey.

11 Aug day 26
★☆☆☆☆ The customer gives 1 star, still waiting.

17 Aug day 32
The parcel finally arrives, 10 days late.

When the parcel arrived, its review was already six days old.

promised date → review

A broken promise turns into a bad review

1–2★ review rate by how late the parcel was · 95,561 delivered and reviewed orders, January 2017 – August 2018

1–2★ REVIEWS

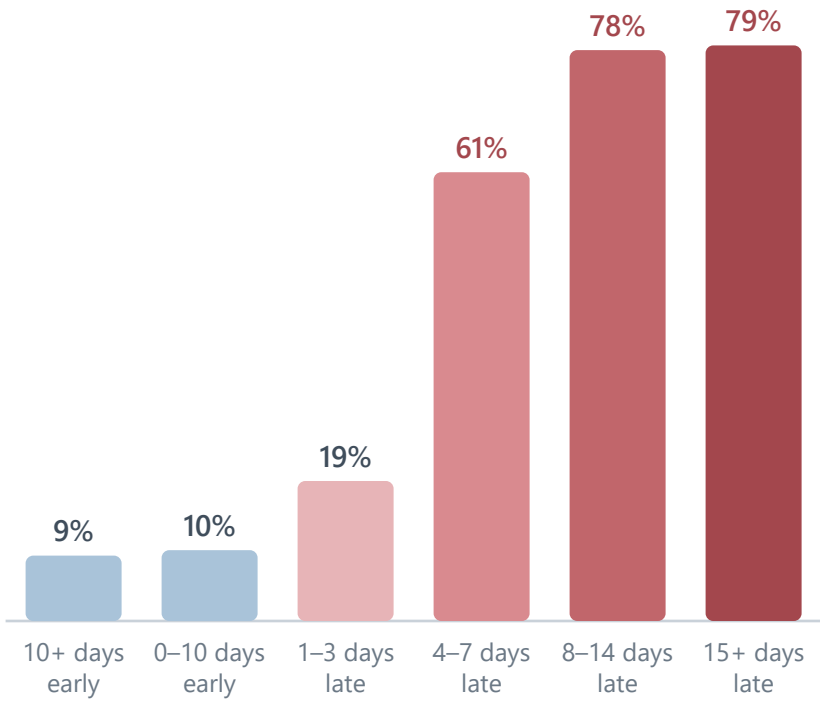
9% parcel on time

54% parcel late

A late parcel makes a 1–2★ review six times more likely — and the later it is, the angrier the review.

Why it matters
Bad ratings cost sellers future sales (Deshpande & Pendem, 2023), and Olist lives on its sellers.

1–2★ review rate, by delivery outcome



WRITTEN BEFORE ARRIVAL

87%

of the bad reviews on late parcels. Olist sends the survey once the promised date passes.

LATE PARCELS

1 in 12

orders arrives late, and produces 34% of all 1–2★ reviews.

Step in before the review — at three moments

Three moments between checkout and the review, each owned by a different team



OPERATIONS

Which promises are going to break?

Quote an honest date. Warn the riskiest buyers early.

uses the delivery forecast

CUSTOMER CARE

The parcel is late, and the survey is about to go out.

Reach the customer first, with an apology and a new date. Ask for the review after the parcel arrives.

needs no model: the calendar

CUSTOMER EXPERIENCE

Who is unhappy even though the parcel came on time?

Two thirds of bad reviews come from on-time parcels, about two days after delivery. Call the riskiest first.

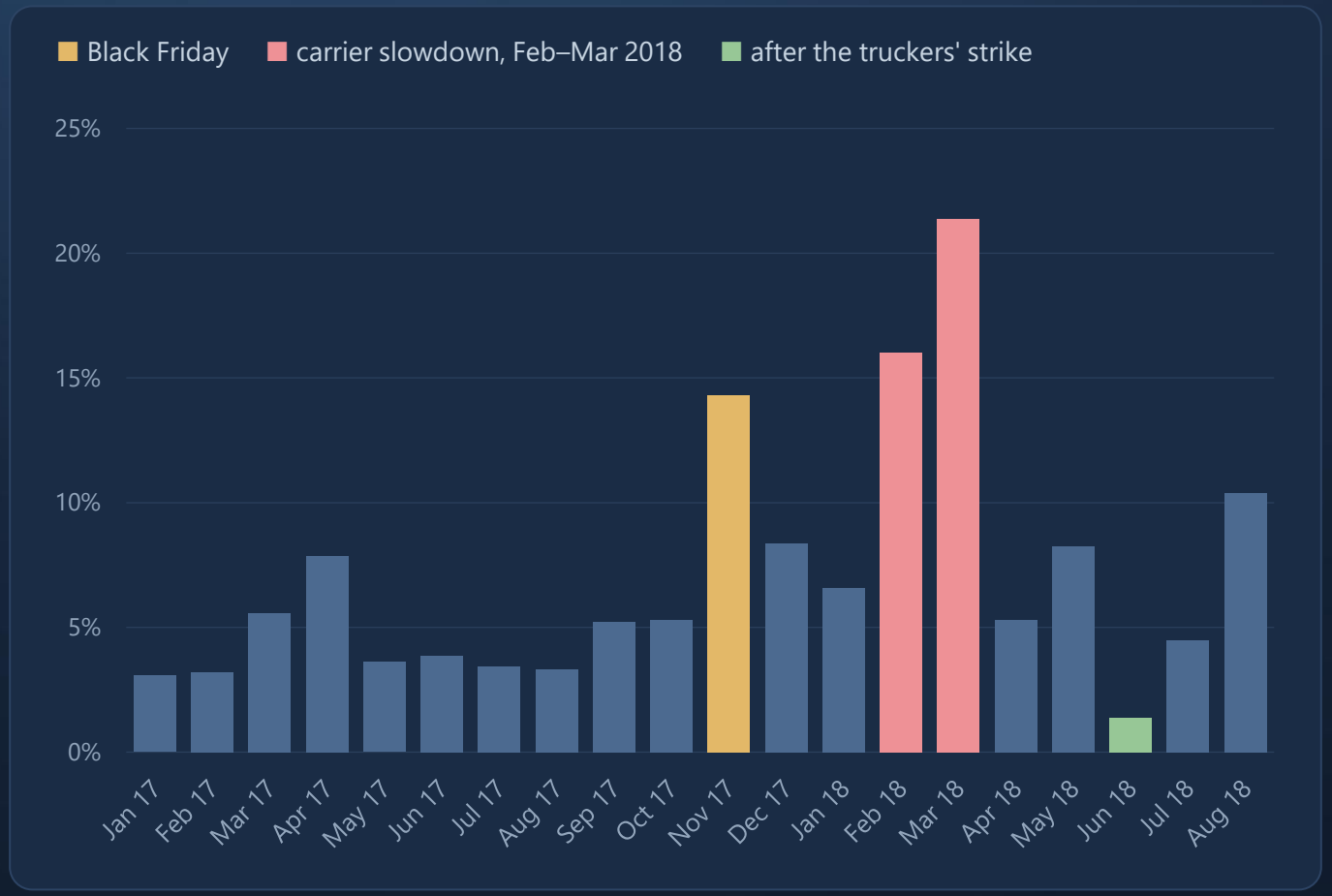
uses the review-risk score

Today Olist finds out here, when it is too late to act.

1 in 8 orders ends in a 1–2★ review

The ground kept moving under the data

Share of parcels delivered late, by month of purchase, January 2017 – August 2018



Twice as fast in six months

Average delivery fell from 17 days (Feb 2018) to 8 (Aug 2018): last year's patterns go stale.

Shocks no single order reveals

Black Friday, a carrier slowdown, a national truckers' strike.

No peeking at the future

One column from the future makes a model perfect (ROC-AUC 1.00) and useless. We trained on 2017 – April 2018 and judged on July – August 2018: months the models never saw.

What we built: an early-warning system

checkout and delivery

Three pieces, each tied to one moment and one team

WHAT GOES IN

99,441 orders from 9 tables

- Promise and distance
- Basket and calendar
- Seller and route records
- How the delivery went

AT CHECKOUT

Delivery forecast

How many days the parcel will take, and the chance the promise breaks.

→ honest date, warning list

AT DELIVERY

Review-risk score

The chance this customer gives one or two stars.

→ ranked call list

FOR THE TEAMS

Live web app

Scores one order or a whole file, replays any test day, and explains itself.

→ team10-it5006-olist.streamlit.app

Under the hood: scikit-learn tree ensembles, picked from three model families on the most recent months. **Every team has good models; the hard part was choosing the moment and the action.**

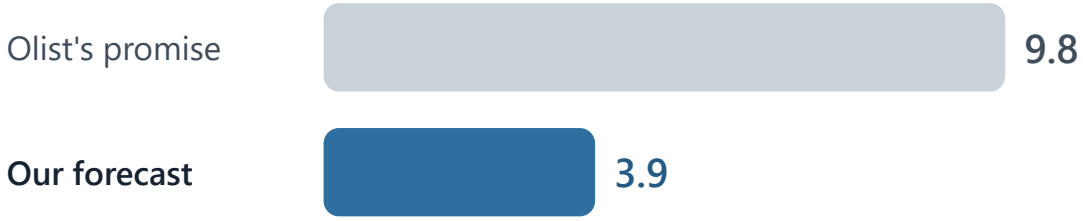


Honest dates, and an early-warning list

Judged on July – August 2018, two months the models never saw

HOW FAR OFF IS THE DELIVERY DATE?

average error, in days



60% closer

Olist pads its promise on purpose. Customers can be shown the honest number instead.

LATE PARCELS IN THE RISKIEST 10% OF ORDERS

35%

3.5× better than picking at random: warn these buyers early. Our São Gonçalo order was #23 of 238 that day.

BUT DON'T PAY TO RUSH THEM

< 30%

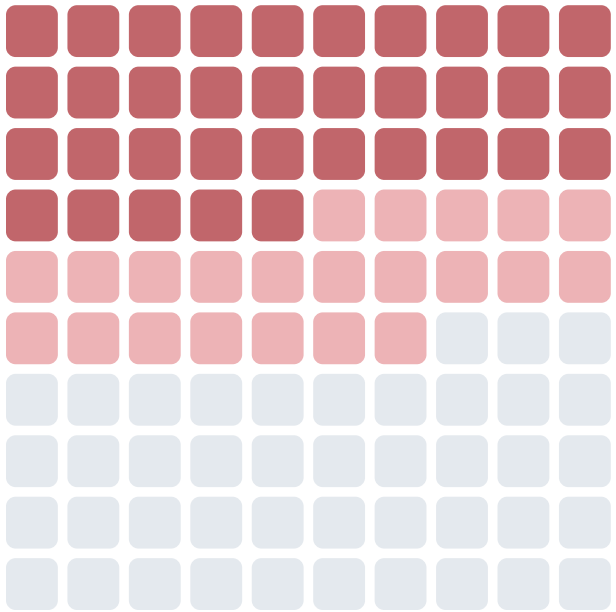
Even the riskiest flagged parcels are late less than 30% of the time. A R$12 expedite needs 30% to pay back.



Ten calls a day find the unhappy customers

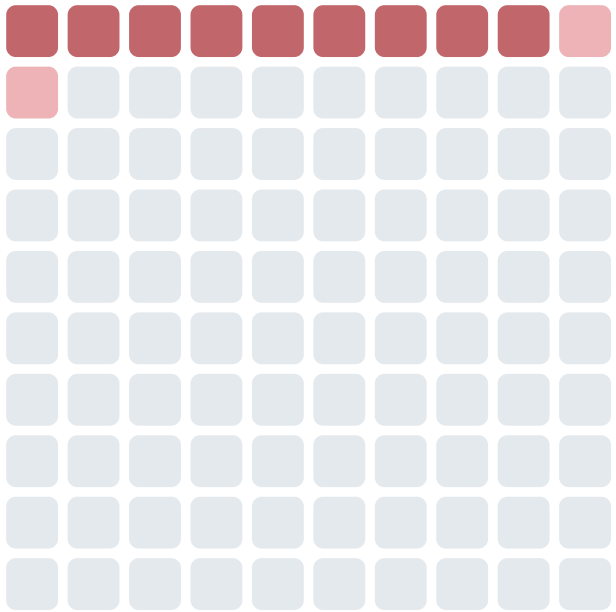
100 calls from our daily top-10 list against 100 random calls, July – August 2018

OUR TOP-10 LIST



57 of 100 calls reach a customer about to give a 1–2★ review

100 RANDOM CALLS



11 of 100 random calls do, by chance alone

■ reached before they write ■ already wrote, waiting for a late parcel ■ no 1–2★ review

BETTER THAN CALLING AT RANDOM

5×

57 against 11 hits per 100 calls.

STILL REACHABLE BEFORE THEY WRITE

35

The other 22 wrote while waiting for a late parcel: for them, the promised date is the moment.

HITS PER 100 CALLS TO BREAK EVEN

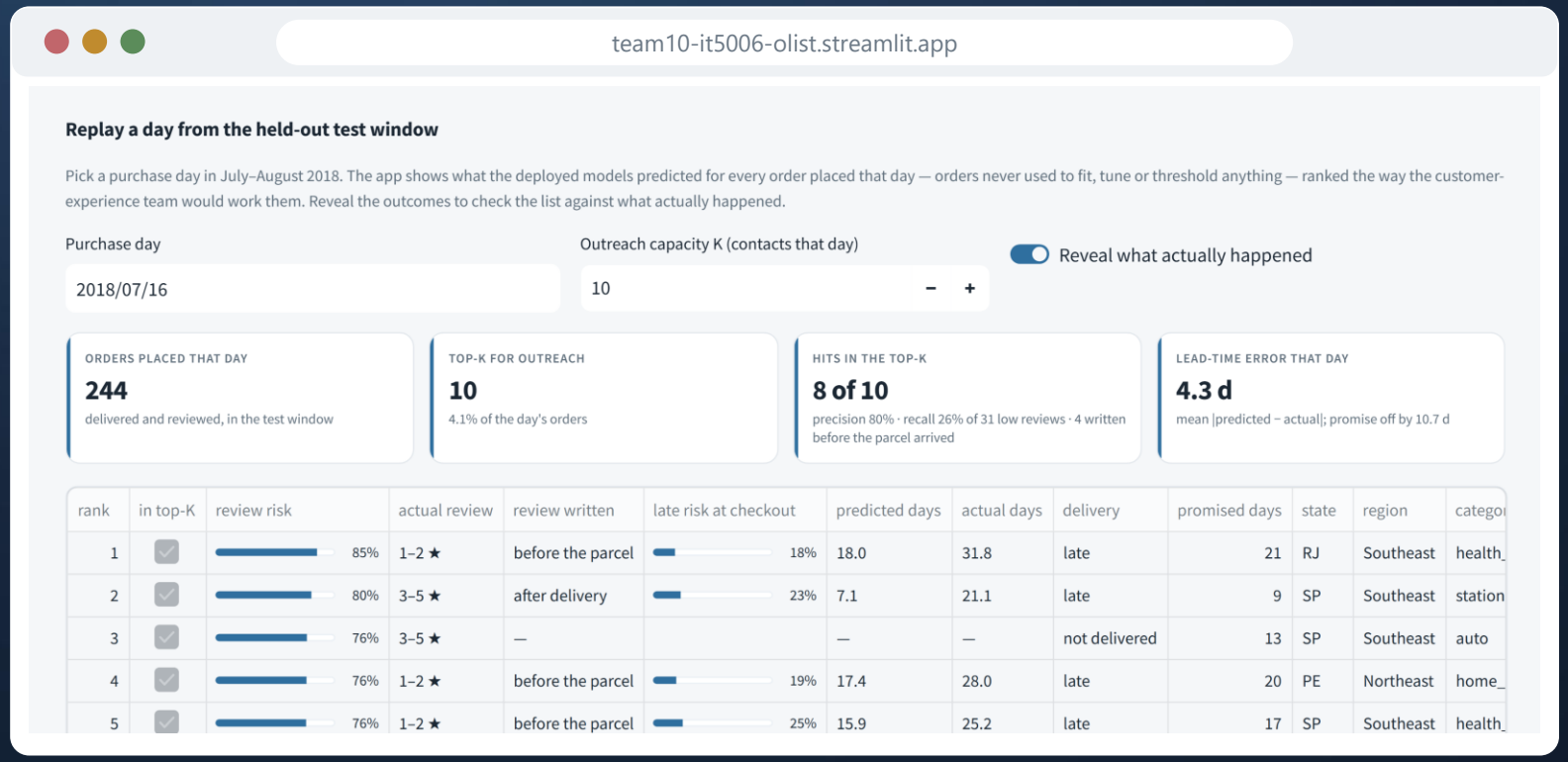
13

A R$5 call against a R$40 bad review. We reach 35.

Replay a day the models never saw

team10-it5006-olist.streamlit.app · Replay a day · 16 July 2018, 244 orders, 10 calls





▶ Open the live app

or scan the code on the last slide

TOP 10 ON 16 JULY

8 of 10

really gave one or two stars

4 still reachable at delivery

4 had already written, still waiting

#1 on the list, at 85% risk: the order we started with. Its review was six days old.

A good day. Across all 62 test days, the list averages 57 hits per 100 calls.



Four changes Olist can make on Monday

Behind all four: refit the models monthly and watch the weekly late rate. The next strike will not announce itself.

OPERATIONS · AT CHECKOUT

Quote the forecast, not the padded promise

Every order. Free. Dates 60% closer to reality.

CUSTOMER CARE · ON THE PROMISED DATE

When a promise breaks, speak first

Apologise with a new date, and send the survey after the parcel arrives. Near-free.

CUSTOMER EXPERIENCE · AT DELIVERY

Call the ten riskiest deliveries every morning

R$5 a call. 57 of 100 calls reach an unhappy customer.

OPERATIONS · WHAT NOT TO DO

Don't pay to expedite on the flag

At R$12 a parcel it never pays back: the flag is right less than 30% of the time.



Up to R$775 per 1,000 orders — if a call helps

Stated assumptions: R$5 per call, R$40 per 1–2★ review. Olist publishes no costs, so we show the arithmetic.

PER 1,000 DELIVERED ORDERS

R$775

at the best call volume, counting only customers still reachable before they write. Ten calls a day: R$433.

THE CATCH

Nobody was ever called

History cannot tell us whether a call changes the review. Only an experiment can.

THE SIX-WEEK EXPERIMENT

1 Every morning: the ten riskiest deliveries



2 Randomly split them



3 After six weeks, compare their 1–2★ rates



Six weeks at ten calls a day detect a one-in-four drop (191 customers per arm, 80% power). Second arm: late parcels, contacted on their promised date.

What we learned

Three things we would tell the next team

01

Timing beats modelling

Every team can fit a good model. Our biggest gain was learning when the review gets written.

02

Judge on the future

Delivery got twice as fast in six months. We chose models on recent months and tested once, on months they never saw.

03

Say what data can't know

Strikes, slowdowns, and whether a call helps. Hence weekly monitoring, monthly refits and the experiment.

Limits we accept: Brazil 2017–18 only · a star rating is not satisfaction · the scores are for triage, never to penalise a seller for a carrier's delay.

THANK YOU

The review arrived before the parcel.

With honest dates, a word before the survey and ten calls a day, Olist can get there first.



Live app

Code, notebooks, report

Pan Pinyou · Li Yudan · Zhu Gengyu — Group 10, IT5006, National University of Singapore

QUESTIONS?

Jump to the detail

- How we tested →
- When reviews are written →
- The models under the hood →
- Checkout numbers →
- Delivery numbers →
- Where the money comes from →
- Limits and next steps →
- The app, tab by tab →

BACKUP · QUESTIONS

How we tested: never on data the model had seen

Each period has one job; the test window was scored once, for models already chosen



Train: fit and tune on five expanding time windows

Validation: choose models and thresholds · Test: scored once, replayed by the app

NO PEEKING AT THE FUTURE

0.589 → 1.000

ROC-AUC of the late-delivery model when one forbidden column (the actual delivery time) slips in. A guard blocks every post-outcome column for each prediction moment.

CUSTOMERS, NOT JUST DATES

227 removed

2,997 people ordered more than once; the 227 who appear on both sides of the split are removed from the clean test subset (the review model scores 0.415 on it, against 0.414 overall).

CHECKED, NOT TRUSTED

60 checks

An automated suite rebuilds the track-record features from first principles, tests the leakage guard both ways and ties the deployed models to the report.

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When is the review written?

Olist's tables carry two review timestamps: when the survey was sent and when it was answered · src/review_timing.py

	Late parcels	On-time parcels
Orders (delivered and reviewed)	7,659 (8%)	87,902 (92%)
Give 1–2★	54.0%	9.2%
Survey sent on or after the promised date	99.0%	2.1%
1–2★ reviews written before the parcel arrived	87.3%	0.5%
Share of all 1–2★ reviews	33.8%	66.2%
Typical 1–2★ review	4.5 days before arrival	2.1 days after delivery

So for late parcels the review usually precedes delivery, and the moment to act is the promised date; for on-time parcels the review follows delivery by about two days, and the call list has time to work.

Call list (test window)	hits	reachable	already written
Top 5% of deliveries	68%	36%	32%
10 calls a day	57%	35%	22%
20 calls a day	44%	29%	15%
Calling at random	11%	9%	2%

VALUE PER 1,000 DELIVERED ORDERS
R$1,466 if every flagged customer could be reached; R$775 counting only those reached before they write (best operating point). Ten calls a day: R$853 and R$433.

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BACKUP · QUESTIONS

The models under the hood

Three model families, each run as a ladder from its simplest version; chosen on the most recent out-of-sample months

A · LINEAR

LinearRegression / LogisticRegression → Ridge / ℓ2 logistic

B · TREES

Decision tree → random forest → HistGradientBoosting

C · ENSEMBLES

Voting and stacking, built from the best of A and B

Problem	Deployed model	Test result
Lead time (checkout)	Voting: linear + HistGradientBoosting, 1:2	MAE 3.90 days (promise 9.79)
Late flag (checkout)	Forecast vs promise, calibrated	ROC-AUC 0.782; 2.89× the floor
Review risk (delivery)	Stacking: ℓ2 logistic + HistGradientBoosting	ROC-AUC 0.749; PR-AUC 0.414 (3.75×)

ONE TREE GETS MOST OF THE WAY

96%

A single depth-limited decision tree reaches PR-AUC 0.396 of the best 0.414.

FOLD-TO-FOLD SPREAD VS THE GAP BETWEEN THE BEST

±0.09 vs <0.001

Cross-validation cannot separate the best trees, so we rank models within each recent window.

WEIGHTING THE LOSS FOR IMBALANCE

risk ×3

No gain in ranking (PR-AUC 0.415 → 0.411); mean predicted risk 0.12 → 0.34. We move the threshold instead.

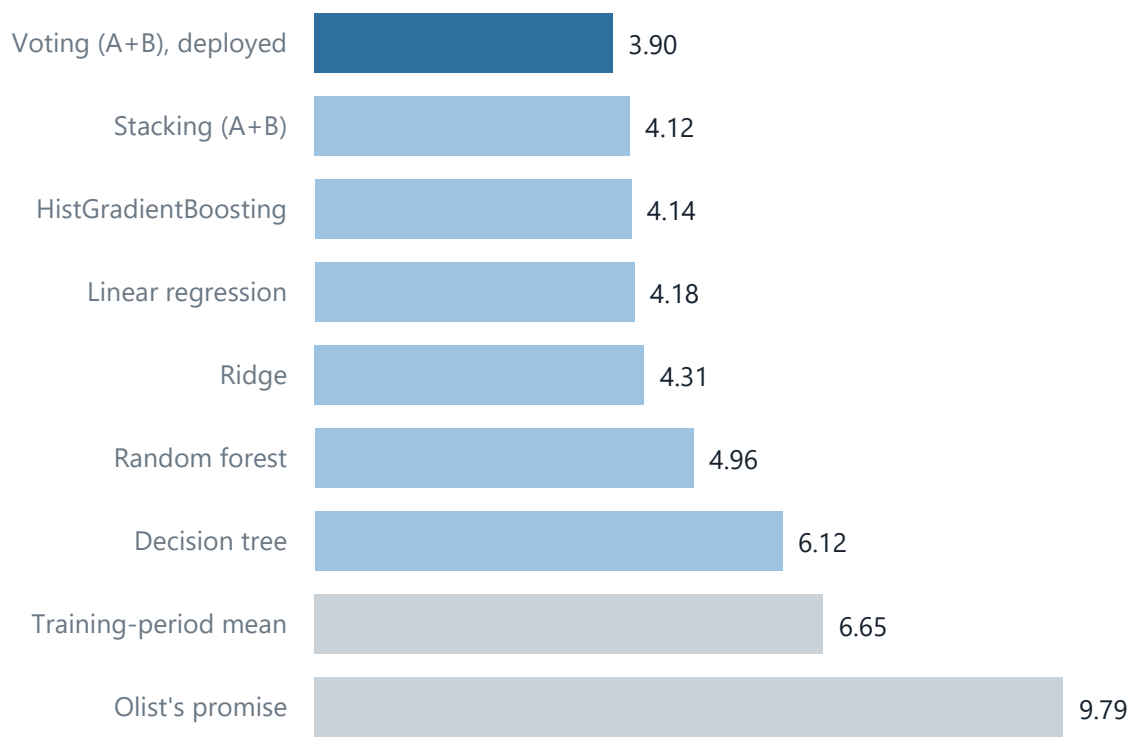
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BACKUP · QUESTIONS

Checkout numbers in full

Lead-time error on the test window for every candidate, and what the late-delivery flag can and cannot do

Test MAE, days (lower is better)



FORECAST VS PROMISE BEATS A DEDICATED CLASSIFIER

2.89× vs 1.24×

PR-AUC lift over the no-skill floor; ROC-AUC 0.782 against 0.584.

WARNING THRESHOLD (11% RISK)

83% caught

...but 45% of buyers warned, 14% of them truly late. The ranked top 10% catches 35% at 26% precision.

EXPEDITING

precision ≤ 0.29

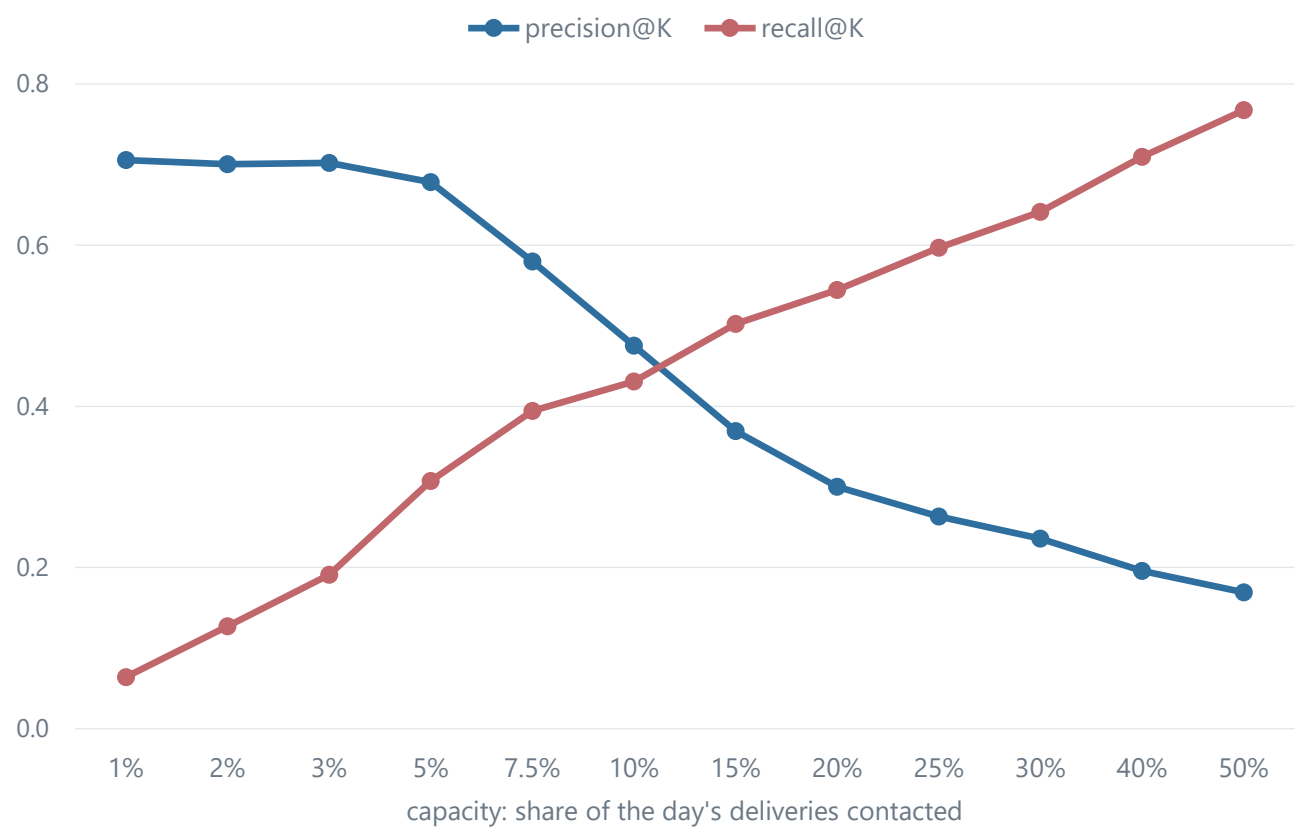
Break-even for a R$12 expedite against R$40 is 0.30; revisit only if expediting costs under about R$11.

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BACKUP · QUESTIONS

Delivery numbers in full

Precision and recall of the review-risk list at every capacity, test window July – August 2018



RANKING QUALITY

ROC-AUC 0.749

PR-AUC 0.414 against an 0.110 floor (3.75×); 0.415 on customers never seen in training.

TOP 5% OF DELIVERIES

68% · 31%

precision and recall; ten a day: 57% and 25%; twenty a day: 44% and 38%.

BREAK-EVEN PRECISION

0.125

R$5 call against R$40 review: cleared at every capacity up to half of all deliveries.

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BACKUP · QUESTIONS

Where the money numbers come from

Costs are stated assumptions, not data: Olist publishes none, so the conclusions rest on break-even precision

Action	Cost of acting	Cost of a miss	Break-even precision	We reach	Verdict
Call before the review	R$5	R$40	0.125	0.35 reachable (0.57 all)	act
Expedite a flagged parcel	R$12	R$40	0.300	≤ 0.29	don't
Quote the honest date	≈ R$0	—	—	60% closer dates	act

PER 1,000 DELIVERED ORDERS

R$775 – R$1,466

best operating point, counting only reachable customers vs counting every flagged one. Ten calls a day: R$433 – R$853.

THE ARITHMETIC

p × R$40 − R$5

per call, where p is the share of calls that reach a would-be 1–2★ reviewer who has not yet written.

WHAT THE PILOT MEASURES

57% → 45%

the cut in the called group's 1–2★ rate that pays back; 191 customers per arm, six weeks at ten calls a day.

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BACKUP · QUESTIONS

Limits, and what we would do next

A moving target

Lead time fell 5.3 days between training and test; the forecast over-predicts everywhere, from +1.2 days (Southeast) to +4.8 (North). Refit monthly.

Shocks no order can see

Black Friday, a carrier slowdown, a truckers' strike: carrier capacity and promise policy, not order attributes. The late flag has a ceiling.

A rating is not satisfaction

Reviews are self-selected; the quietly unhappy customer never writes one. Product, price and seller effects are not in the data.

Triage, not judgement

The review score largely restates lateness: never use it to penalise a seller for a carrier's delay.

NEXT

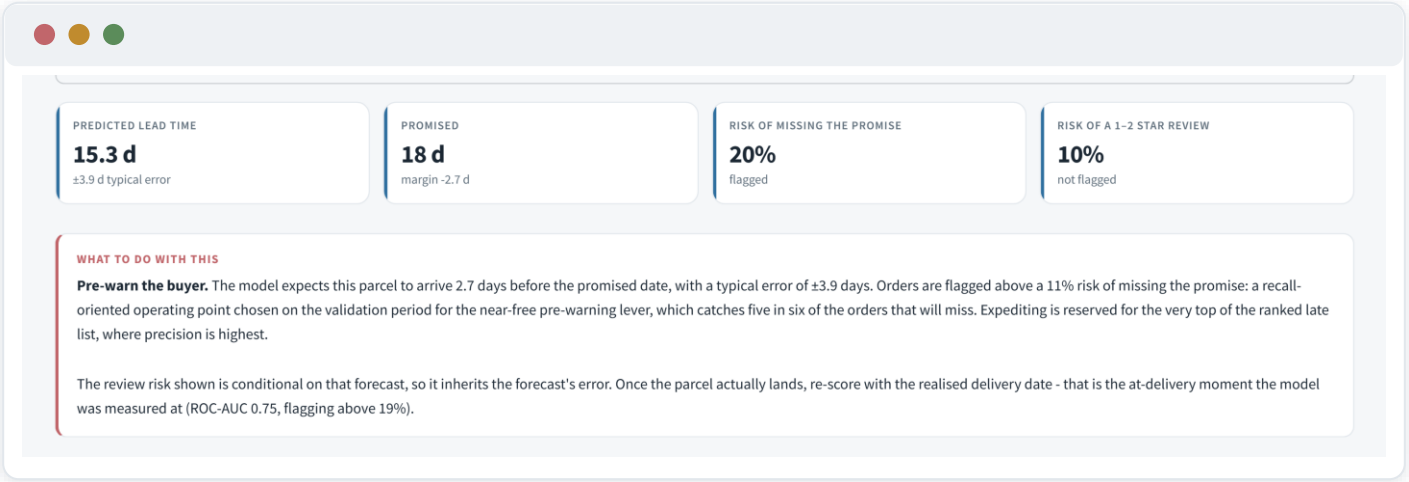
- Run the six-week pilot (two arms: calls at delivery, contact on the promised date).
- Test sending the survey after the parcel arrives.
- Add carrier tracking data; refit weekly during peaks such as Black Friday.

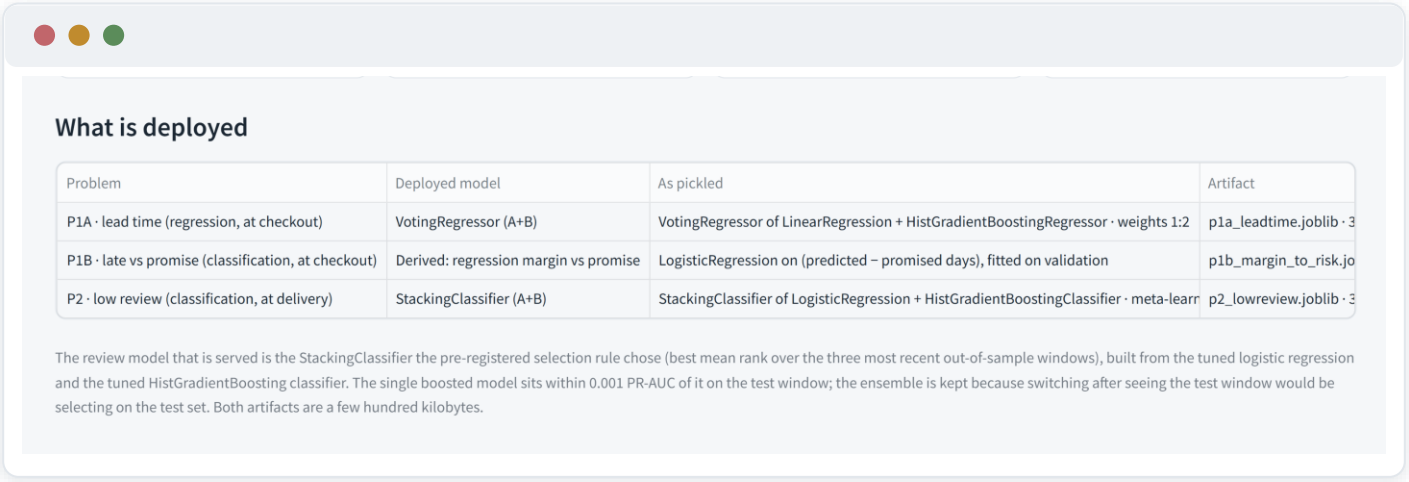
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BACKUP · QUESTIONS

The app behind the story

team10-it5006-olist.streamlit.app · the two screens a team uses besides Replay a day (slide 9)





PREDICT · ONE ORDER AT CHECKOUT

Enter an order: the app returns the forecast delivery days, the risk of missing the promise and what operations should do with it.

MODEL CARD · WHAT IS DEPLOYED

The three deployed estimators, their test metrics and fold-to-fold spread, where not to trust them, and a calculator for the pilot.

ALSO IN THE APP

Batch scoring of a CSV file into a ranked call list (in Predict), Replay a day, and five tabs exploring the data: Overview, Delivery, Geography, Satisfaction, Data quality.

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